# FTDN: Multispectral and Hyperspectral Image Fusion With Diverse Temporal Difference Spans

Xu Chen, Xiangchao Meng, *Member, IEEE*, Qiang Liu, Huiping Jiang, Gang Yang, *Member, IEEE*, Weiwei Sun, *Senior Member, IEEE*, and Feng Shao, *Member, IEEE*

***Abstract*— Multispectral (MS)–hyperspectral (HS) image fusion, which aims to enhance the spatial resolution of low-spatial-resolution HS images with a high-spatial-resolution MS, has provided a wide range of applications in remote sensing. However, relatively long revisit cycles of HS satellites and irresistible weather factors cause the acquisition of HS and MS images at the same time difficult. Most of the existing approaches neglect the temporal difference between MS and HS images and perform weakness in the challenging case with diverse temporal difference spans. In this article, we propose a novel image fusion strategy with embedding a stage of feature matching before interaction. On the one hand, we explore the role of spectral correlation modeling between HS and MS images, which accounts for the utilization of available spatial information from MS images. On the other hand, we design a feature aggregation module to fully exploit the nonlinear gaps and dependencies of heterogeneous data and utilize adaptive gains to realize complementary information projection and fusion. We build Dongying (DY) and Yellow River Estuary (YRE) remote-sensing datasets based on Sentinel-2 and ZiYuan (ZY)-1 02D satellites with diverse temporal difference spans. The extensive experiments demonstrate that our method is robust to the span of temporal difference and shows superior performance over the existing methods visually and quantitatively.**

***Index Terms*— Hyperspectral (HS) image, multispectral (MS) image, remote sensing, spatial–spectral fusion, temporal change.**

Manuscript received 26 March 2023; revised 20 May 2023 and 17 June 2023; accepted 29 June 2023. Date of publication 14 July 2023; date of current version 27 July 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 42171326 and Grant 62071261, in part by the Zhejiang Provincial Natural Science Foundation of China under Grant LR23D010001 and Grant LY22F010014, in part by the Postdoctoral Research Foundation of China under Grant 2020M672490, in part by the Ningbo Natural Science Foundation under Grant 2022J076, in part by the 2025 Science and Technology Major Project of Ningbo City under Grant 2021Z107 and Grant 2022Z032, and in part by the Zhejiang Province "Pioneering Soldier" and "Leading Goose" research Project under Grant 2023C01027. *(Corresponding authors: Xiangchao Meng; Weiwei Sun.)*

Xu Chen, Xiangchao Meng, Qiang Liu, and Feng Shao are with the Faculty of Electrical Engineering and Computer Science, Ningbo University, Ningbo 315211, China (e-mail: chenxu5251@163.com; mengxiangchao@nbu.edu.cn; qiangliu0722@163.com; shaofeng@nbu.edu.cn).

Huiping Jiang is with the Key Laboratory of Regional Sustainable Development Modeling, Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences, Beijing 100101, China, and also with the International Research Center of Big Data for Sustainable Development Goals, Beijing 100094, China.

Gang Yang and Weiwei Sun are with the Department of Geography and Spatial Information Techniques, Ningbo University, Ningbo 315211, China (e-mail: yanggang@nbu.edu.cn; sunweiwei@nbu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3294347

## I. Introduction

REMOTE-SENSING images with both high spatial and high spectral resolutions are widely demanded in various fields, such as change detection [1], target tracking [2], land cover classification [3], and image interpretation [4]. However, constrained by the hardware and imaging systems of satellite sensors [5], current satellite sensors have to make a fundamental tradeoff between spectral and spatial resolutions. For example, Ziyuan (ZY)-1 02D satellite provides hyperspectral (HS) images with high spectral resolution (166 spectral bands) but a relatively low spatial resolution (30 m). Sentinel-2 satellite provides multispectral (MS) images with a high spatial resolution (10 m) but a low spectral resolution (13 spectral bands). HS/MS image fusion is an effective way to produce images with high spectral and spatial resolutions to overcome the inherent hardware limitations.

For multisource remote-sensing images, consistency of the acquisition time is critical, while different acquisition times and conditions may bring about changes in land cover and seasonal spectral variations [6]. The HS images with long intervals are usually unavailable at specified times while unsatisfactory cloud pollution and noise will also reduce the value of data. Borsoi et al. [7] proposed to add a matrix of factors to modify the seasonal spectral variations for MS–HS fusion. Furthermore, Borsoi et al. [8] took into account the local spatial and spectral changes, and an additive variation term was admitted for MS/HS fusion model. However, the above-mentioned work involving temporal differences did not fully model nonlinear spatial–temporal changes, especially for land-cover changes. The deep-learning-based method with strong nonlinear expressiveness has more advantages in this respect.

The existing MS–HS fusion methods were generally founded on two strategies including [9]: early fusion and late fusion. As shown in Fig. 1(a) and (b), the former completes the preliminary fusion before feeding the features into the network, while the latter first transforms the multimodality signals to the feature domain and then conducts deep interaction of multimodality features. However, such strategies may lose efficacy in the face of large modality differences, such as MS and HS observations with diverse temporal difference spans.

As shown in Fig. 1(c), we embed a novel feature matching module before fusion as a vinculum by fully exploiting the

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:05:10 UTC from IEEE Xplore. Restrictions apply.

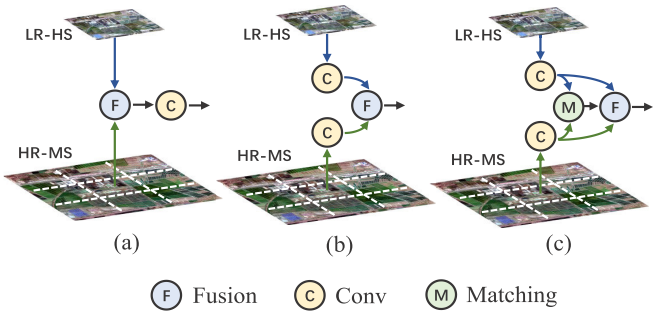



Fig. 1. Examples of the strategies of image fusion. Rather than previous approaches, we regard feature matching as a necessary and explicit stage before feature fusion. (a) Early fusion. (b) Late fusion. (c) Proposed.

commonalities and the differences between the observations and providing rational guidance for further fusion.

In this work, we propose a novel dual-stream network for MS–HS image fusion with diverse temporal difference spans (FTDN). Specifically, three successive modules are proposed for interaction and optimization. First, heterogeneous feature extractors (HFEs) are developed to expand MS and HS images into individual modality features for further application and mutual supplementation. Then, the pixel-wise relevance of spectral curves between MS and HS images is acquired by the spectral correlation matching (SCM) module to exploit common information from reference MS images as valuable knowledge. Correspondingly, a couple of symmetrical feature aggregation (SAF) modules are developed to integrate spatial and spectral information, which utilizes the covariance to regulate the proportion of exchanged information and supervises the adaptive fusion of features. We assume three temporal differences cases from short to long, termed Case 1, Case 2, and Case 3, while the first two cases are practical, and the latter is an extreme situation that is only discussed in experiments. Extensive experiments conducted on Dongying (DY) and Yellow River Estuary (YRE) datasets prove the superiority of our method among the state-of-the-art. The main contributions of our work are summarized as follows.

1) We propose a novel image fusion strategy with an explicit feature matching stage to tackle the problem of potential acquisition temporal difference, which can be applicable to diverse temporal difference spans even in the same acquisition time.
2) We design an SCM module for common information capture and feature discrepancy mitigation and an SFA module to utilize the covariance to regulate the proportion of spatial and spectral information for complementary feature fusion.
3) Extensive experimental results with diverse temporal difference spans conducted on DY and YRE datasets demonstrate that the proposed method achieves state-of-the-art performance with great generalization capability.

## II. Related Work

### A. Spatial–Spectral Fusion

The research on MS–HS spatial–spectral fusion has become more mature during the past decade [10], [11], [12], [13], [14], which can be classified into the following categories [15]: 1) component substitution (CS)-based methods; 2) multiresolution analysis (MRA)-based methods; 3) optimization-based methods; 4) unmixing-based methods; and 5) deep-learning (DL)-based methods. The CS-based method replaces the intensity ("spatial") component of the HS image with a high-spatial-resolution MS image to achieve spatial enhancement of the HS image, such as Principal Component Analysis (PCA) [16] and Gram–Schmidt adaptive (GSA) methods [17]. The MRA-based method extracts high-frequency spatial information from MS images, and they are then injected into the upsampled HS image, such as generalized Laplacian pyramid (GLP) [18] and additive wavelet luminance proportional (AWLP) [19]. CS- and MRA-based methods are generally skilled in spatial enhancement and spectral fidelity, respectively, and do not fully consider the relations of space and spectrum. The optimization-based method constructs an optimal fusion model to adequately describe the relations between observed and desired images [20], [21]. However, it generally acquires relatively higher complexity and time consumption to achieve an acceptable result. The unmixing-based method is based on the linear mixing spectral model (LMM) [22], which regards each coarse pixel of HS images as a mixture of pure endmembers of MS images and can obtain a positive spectral quality. The typical methods include the HS image super-resolution method (Hysure) [23] and coupled nonnegative matrix factorization (CNMF) [24]. The DL-based algorithm has been introduced for remote-sensing image fusion and achieved more remarkable results than classical methods [25], [26], [27], [28]. Masi et al. [29] first applied a three-layer convolutional neural network (PNN) to pansharpening and inspired future work. Xie et al. [30] proposed an interpretable deep network (MHF-Net) for MS–HS fusion based on the low rank of HS prior knowledge. Zhang et al. [31] adopted a lightweight spatial–spectral reconstruction network (SSRNet) for effective MS–HS image fusion and get an impressive result at a small parameter cost. Hu et al. [32] proposed an HS image super-resolution network (HSRnet) with the applications of the pixelShuffle and attention modules to keep high-quality spatial details preserved.

### B. Spatial–Temporal–Spectral Fusion

Spatial–spectral fusion is dedicated to fusing the spatial and spectral resolution indexes, and some integrated spatial–temporal–spectral methods are proposed to incorporate time relationships into the discussion. Li et al. [6] proposed an integrated framework for the spatial–temporal–spectral fusion of remote-sensing images, which completes multiple kinds of fusion tasks with an arbitrary number of satellite data. Zhao et al. [33] proposed a sparse representation model to utilize available heterogeneous spatial, spectral, and temporal data to generate synthesized frequent high spatial and spectral resolution data. Peng et al. [34] proposed a spatial–temporal–spectral fusion model based on semicoupled sparse tensor factorization, which decomposes this problem into the estimation of core tensors and dictionaries along temporal, spatial, and spectral modes. Zhou et al. [35] proposed a generalized linear spectral mixing model for spatial–temporal–spectral

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:05:10 UTC from IEEE Xplore. Restrictions apply.

fusion and captured the endmember variations at adjacent time through coefficient factors to represent the linear spectral variations of observations. Liu et al. [36] proposed a dual-task learning model based on deep learning for spatial–temporal–spectral fusion, which solves the complex nonlinear relationships between multisource observations profiting by the modeling and interaction of temporal and spectral relationship branches.

However, the aforementioned spatial–spectral methods appear to be struggling with different imaging times [37], while the implementation of spatial–temporal–spectral fusion requires at least two MS/HS image pairs. The proposed FTDN is committed to solving the imaging temporal difference in a general model and achieving more competitive results than the previous methods.

## III. Proposed Method

In this section, the framework of FTDN is built on a dual-stream network that adopts successive MS and HS coupling iteration layers (MHCIL) for learning, mainly including an SCM module and a couple of SFA modules.

The details of SCM and SFA will be discussed in Section III-A and the loss functions will be elaborated in Section III-B.

### A. General Framework

The framework of FTDN is shown in Fig. 2. Specifically, we should clarify the symbols frequently appearing in this article, HS $\in \mathbb{R}^{h\times w\times C}$ and MS $\in \mathbb{R}^{H\times W\times c}$ refer to input HS and MS images. HS $\uparrow\in \mathbb{R}^{H\times W\times C}$ is derived from bicubic upsampled HS. $F \in \mathbb{R}^{H\times W\times C}$ and $X \in \mathbb{R}^{H\times W\times C}$ refer to the fused image and the reference, respectively. $H$ and $W$ denote spatial dimensions, and $C$ denotes the spectral bands ($H \gg h$, $W \gg w$, and $C \gg c$).

### B. Heterogeneous Feature Extractor

A pair of HFE is applied to extract multilevel features from MS and HS images, and learnable parameters are continuously updated during network training. For the reliability of matching, the extractor has consistency in structure to extract equivalent semantic features, where two adjacent convolution blocks with the kernel size of $3 \times 3$ project the hierarchical feature maps into the shape of $H \times W \times B$. The process of HFE is written as follows:

$$M_1, M_2, \ldots, M_n = \mathrm{HFE}_m(M_0) \tag{1}$$

$$H_1, H_2, \ldots, H_n = \mathrm{HFE}_h(H_0) \tag{2}$$

where $\mathrm{HFE}_m(\cdot)$ and $\mathrm{HFE}_h(\cdot)$ denote the operations of HFE. $M_0$ and $H_0$ are obtained from the single $3 \times 3$ convolution layer. $M_1, M_2, \ldots, M_n$ and $H_1, H_2, \ldots, H_n$ denote hierarchical feature maps, and the above feature will be fed into SCM consequently.

### C. Spectral Correlation Matching Module

SCM aims to quantify the contributions of spatial texture in MS images based on the spectral correlation between MS and HS image pairs pixel-for-pixel.

MS–HS image fusion suffers from various restrictions, including modality difference and potential temporal difference, which introduce invalid information from MS image to the fusion model and bring negative misleading or noise, especially in complex object prediction. Inspired by Texture Transformer Network [38], spectral correlation provides an intuitive way to get aware of common contents between MS and HS image pairs and assigns attention to each pixel according to the similarity.

Citing the case of the $n$th layer in MHCIL, $M_n$ and $H_n$ are projected into the common feature domain through linear mapping of shared weights for comparison. Given the presence of spectrum mixing in low-resolution (LR) HS pixels, the spatial resampling operation can ensure that the internal spectral compositions of HS and MS features are equivalent. Moreover, $l_1$-norm is used to facilitate further data analysis, the process is depicted as

$$M'_n = \|\mathrm{resample}(\mathrm{Conv}_{1\times1}(M_n))\|_1 \tag{3}$$

$$H'_n = \|\mathrm{Conv}_{1\times1}(H_n)\|_1 \tag{4}$$

where $\mathrm{conv}_{1\times1}(\cdot)$ indicates the linear mapping. With $M'_n$ and $H'_n$, the correlation coefficient $m$ is depicted as

$$m_{i,j} = \frac{\tilde{H}_{i,j} \cdot \tilde{M}_{i,j}}{\left\|\tilde{H}_{i,j}\right\|_2^2 \left\|\tilde{M}_{i,j}\right\|_2^2} \tag{5}$$

where $\tilde{M}_{i,j}$ and $\tilde{H}_{i,j}$ mean the spectral vector at pixel $(i, j)$. $m \in \mathbb{R}^{H\times W}$ is fine-tuned by convolution and mapped into $(0, 1)$ through Sigmoid activation. $M_n$ and $H_n$ are supposed to be distinctive in weight allocation to boost active fusion of the regions with strong relevance or preserve original HS information instead. Finally, $G_n$ is derived from the weighted averaging of $M_n$ and $H_n$ with weights $h$ and 1-$h$, and the process is depicted as follows:

$$h = \mathrm{Sigmoid}(\mathrm{Conv}_{1\times1}(m)) \tag{6}$$

$$G_n = (h \odot M_n) \oplus ((1 - h) \odot H_n) \tag{7}$$

where $G_n$ is regarded as a linkage of MS and HS features and further provides matched information for SFA.

### D. Symmetrical Feature Aggregation Module

To realize MS and HS feature deep coupling, SFA is applied to refine the initial features of $M_n$ and $H_n$ with matched information, which injects the details from $G_n$ in an adaptive manner.

The common feature fusion approaches, for instance, concatenation, element/point-wise addition, and attention-based operations, however, are suffering from effect constraints or information loss [39]. Inspired by the probability theory, here takes variance and covariance into account, where variance measures the dispersion degree of variables, and covariance can be used for dependence calculation between variables,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:05:10 UTC from IEEE Xplore. Restrictions apply.

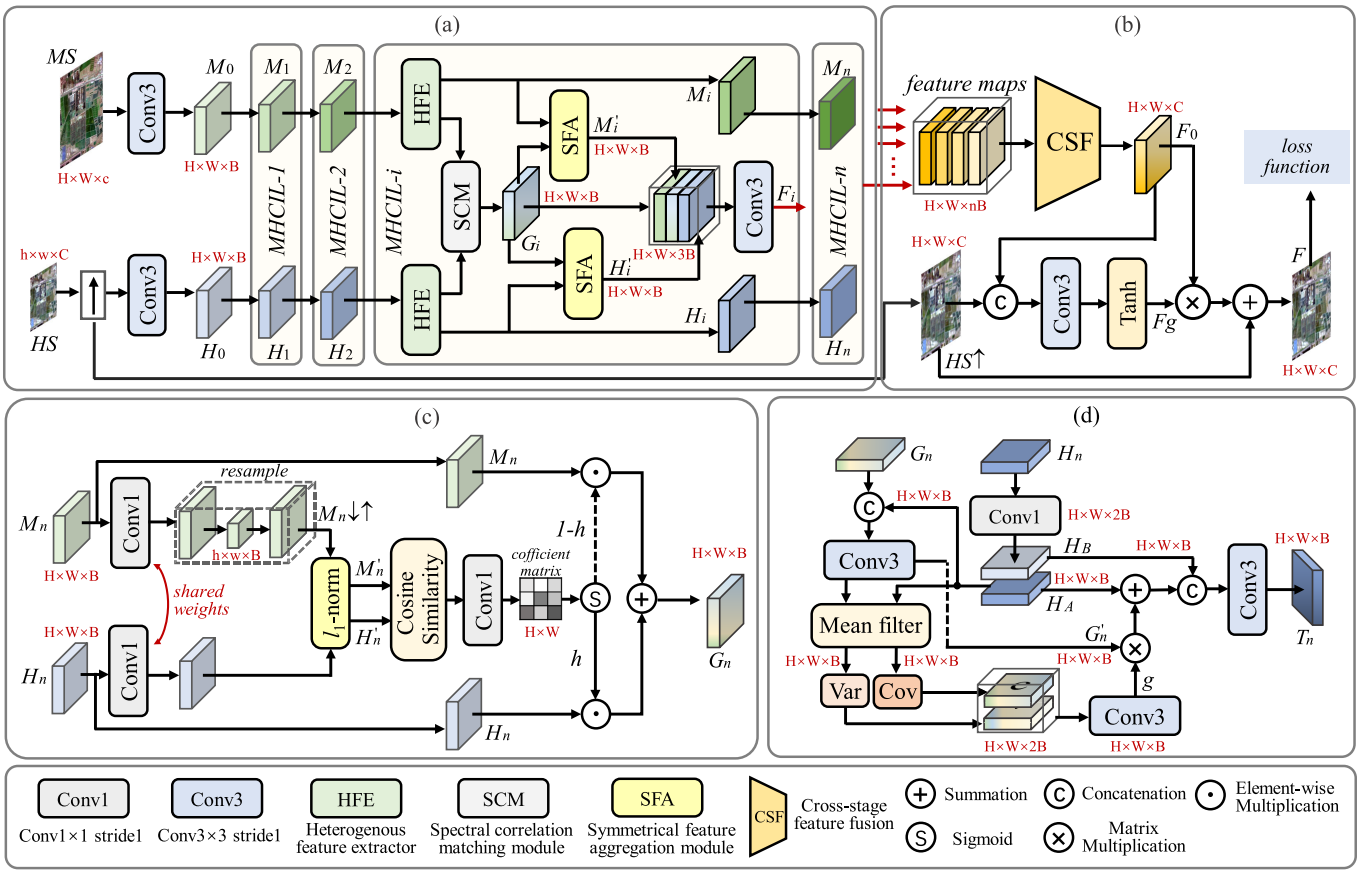



Fig. 2. Flowchart of the proposed model. From (a) to (d) are the details of MHCIL, IAR, an SCM module, and an SFA module.

to acquire the gains of exchanged information for effective feature fusion.

Take the HS branch as an example, first, the channels of $H_n$ are doubled and divided into two components of $H_A$ and $H_B$ with the same dimension of $H \times W \times B$. Besides, $H_B$ is retained as original information, and it is depicted as

$$H_A, H_B = \text{Split}(\text{Conv}_{1\times1}(H_n)). \tag{8}$$

Second, $G_n$ and $H_A$ are cascaded along the spectral dimension and fed into the convolution block with a kernel size of $3 \times 3$ to acquire the general knowledge of input features as

$$G_n' = \text{Conv}_{3\times3}(\text{Concat}(G_n, H_A)) \tag{9}$$

and the calculation process can be depicted as

$$\text{Cov}\left(H_A, G_n'\right) = E\left(H_A \cdot G_n'\right) - E(H_A) \cdot E\left(G_n'\right) \tag{10}$$

$$\text{Var}\left(G_n'\right) = E\left(G_n' \cdot G_n'\right) - E\left(G_n'\right) \cdot E\left(G_n'\right) \tag{11}$$

where $E(\cdot)$ refers to the mean value, and $\text{Cov}(\cdot\,,\cdot)$ and $\text{Var}(\cdot)$ refer to covariance and variance, respectively. Later, the combination of $\text{Cov}(H_A, G_n')$ and $\text{Var}(G_n')$ is convolved to generate the revised gain $g$ for the enhancement of spatial details, and it is depicted as follows:

$$g = \text{conv}_{3\times3}\left(\text{Concat}\left(\text{Cov}\left(H_A, G_n'\right), \text{Var}\left(G_n'\right)\right)\right) \tag{12}$$

where $g$ is the generated gain. After that, the input $G_n'$ is regulated by the gains $g$ to enhance the information of interests. We merge the enhanced $H_A$ and original $H_B$ to strike a balance in sharpening and fidelity of features as

$$H_n' = \text{conv}_{3\times3}\left(\text{Concat}\left(\left(g \otimes G_n' \oplus H_A\right), H_B\right)\right) \tag{13}$$

where $H_n'$ refers to the aggregation feature. The process is also adopted by the MS branch. Finally, the complementary output of MS and HS branches in SFA are combined to get an intral-stage result.

### E. Image Adaptive Reconstruction

The cross-stage features $F_1, F_2, \ldots, F_n$ of MHCIL are further stacked in the cross-stage fusion (CSF) module to generate an integral $F_0$, which enriches semantic information to the basic features. For the exchange of messages between $F_0$ and source image HS $\uparrow$, we introduce an adaptive soft gating as an excitation of cooperation or competition in the feature domain. Specifically, HS $\uparrow$ and $F_0$ are combined and convolved with the kernel size of $3 \times 3$ to strengthen the cognition between features, and it is depicted as follows:

$$F_g = \text{Tanh}(\text{Conv}_{3\times3}(\text{Concat}(\text{HS}\uparrow, F_o))) \tag{14}$$

where Tanh nonlinear activation supersedes Sigmoid for pixel value updating, which simultaneously promotes the convergence of the network. The weighted features are further integrated with HS $\uparrow$ as follows:

$$F = (F_g \otimes F_o) \oplus \text{HS}\uparrow. \tag{15}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:05:10 UTC from IEEE Xplore. Restrictions apply.

TABLE I
SPECIFIC CONTENT OF THE DY DATASET AND THE YRE DATASET

| Dataset | Settings | Sentinel-2(MS) | ZY1-02D(HS) |
|---|---|---|---|
| DY | Acquisition time | 2019.12.08 | 2019.12.11 |
| | | 2020.01.02 | |
| | | 2021.02.10 | |
| | Spatial resolution | 10m | 30m |
| | Bands | 4 | 151 |
| | Coverage range | 0.4μm-2.4μm | 0.4μm-2.5μm |
| | Training set | 72×72×4000p | 24×24×4000p |
| YRE | Acquisition time | 2019.12.08 | 2019.12.11 |
| | | 2021.02.10 | |
| | Spatial resolution | 10m | 30m |
| | Bands | 4 | 151 |
| | Coverage range | 0.4μm-2.4μm | 0.4μm-2.5μm |
| | Training set | 72×72×1500p | 24×24×1500p |

Finally, we obtain the reconstruction image of the MS–HS image fusion task.

### F. Loss Function

The overall loss function consists of two parts, including general reconstruction loss $l_1$ and spectral fidelity loss $l_{\text{spe}}$, the loss is depicted as

$$\text{loss} = \lambda_1 l_1 + \lambda_2 l_{\text{spe}} \tag{16}$$

where $\lambda_1$ and $\lambda_2$ denote the weight of loss function.

*1) General Reconstruction Loss:* $l_1$ computes the mean absolute errors (MAE) of fusion images and HR-HS image, which can be less sensitive to the outliers than mean square error (mse) and shows the superiority of spatial edge, and it is defined as

$$l_1(X, F) = \|X - F\|_1. \tag{17}$$

*2) Spectral Fidelity Loss:* penalization term $l_{\text{spe}}$ is introduced to measure the spectral distortion between fusion images and HR–HS images, and it is defined as

$$l_{\text{spe}}(X, F) = \frac{1}{HW} \sum_{h,w} \arccos\left( \frac{\tilde{X}_{h,w} \cdot \tilde{F}_{h,w}}{\left\|\tilde{X}_{h,w}\right\|_2 \left\|\tilde{F}_{h,w}\right\|_2} \right). \tag{18}$$

Empirically, the weights of $\lambda_1$ and $\lambda_2$ are set to 1 and $1 \times e^{-1}$, respectively.

## IV. EXPERIMENTS

### A. Experimental Settings

Based on the point of departure of this article, we have considered the MS–HS image fusion under the following temporal difference spans.

1) Case 1 with consistent acquisition time.
2) Case 2 with a slight temporal difference.
3) Case 3 with a large temporal difference.

Consequently, two multitemporal datasets of DY and YRE comprised of Sentinel-2 MS image sequence and ZY-1 02D HS image are employed to validate our method.

In data processing, the raw MS and HS sacrificed invalid bands and were cropped into patches with the size of 72 × 72 and 24 × 24, respectively. Under Wald's protocol [41], the patches were blurred by a 7 × 7 Gaussian filter with a variance of 5 and decimated by the factor of 3 as input data, and the original HS patches are taken as references. Thus, for the DY dataset, the training set contains 4000 pairs of MS–HS patches, while there are 1500 pairs for the YRE dataset. Table I explains the details.

To perform the objective fusion quality assessment of the simulated and real data, five common metrics of the peak signal-to-noise ratio (PSNR), structural similarity (SSIM), spectral angle mapper (SAM), and spatial correlation coefficient (SCC) are adapted for evaluation on simulated data, and three nonreference metrics of $D_s$, $D_\lambda$, and QNR for evaluation on real data. The indices PSNR and SSIM are widely applied to make a comprehensive evaluation when SAM and SCC focus on the measurement of spectral distortion and spatial quality, respectively. In addition, some well-approved methods for MS-HS image fusion including traditional methods of Hysure [24], a fast fusion of multiband images based on solving a Sylvester equation (FUSE) [41], GSA [17], and HS–MS image fusion with spectral variability (Fuvar) [7], and DL-based methods of SSRNet [31], ResTFNet [42], MHF-Net [30], and HSRnet [32] are used for comparison.

For the network implementation, FTDN is conducted on PyTorch based on NVIDIA RTX A6000, and the parameters in our network were updated by the Adam optimizer. The batch size is set to 8 and the learning rate to $1 \times e^{-1}$ with 0.5 decay per 100 epochs. The final model got from near $1 \times e^4$ interactions achieves the best comprehensive performance.

### B. Comparison on Reduced-Resolution Scene

In this section, extensive experiments corresponding to the two datasets discuss the three fusion cases, and qualitative and quantitative evaluations will examine the performance of all comparison methods.

*1) Case 1:* Table II shows quantitative evaluations of the DY dataset, where the bold letters signify the best and the underlined letters signify the second best performance. Generally, DL-based methods achieve better effects than the traditional methods, while FTDN gets more superior indicators in DY datasets. The qualitative evaluations are shown in Fig. 3 for intuitive visual comparison. The traditional methods are weak in spectral expression and DL-based methods show apparent blur in object edge. Our proposed method gains satisfactory spatial texture with less perceptible distortion and shows a more competitive effect among all the methods. To validate the generalization ability in various datasets, we also train the model on the YRE dataset. The quantitative indicators are shown in Table III. As can be seen, PSNR and SCC in FTDN are of great advantages, whereas other methods show performance fluctuations across the datasets. Thus, FTDN shows a stronger adaptability relative to other competitors.

*2) Case 2:* Temporal difference of less than a month commonly shows slight varieties in vegetation and water. The quantitative results of the comparison methods are shown in Table II. As can be seen, all indicators have declined

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:05:10 UTC from IEEE Xplore. Restrictions apply.

TABLE II

QUANTITATIVE INDICATORS OF IMAGE FUSION OF THREE CASES ON THE DY DATASET. (THE BEST IS MARKED IN BOLD AND THE SECOND BEST IS UNDERLINED)

| | Case1 | | | | Case2 | | | | Case3 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Methods | PSNR | SSIM | SAM | SCC | PSNR | SSIM | SAM | SCC | PSNR | SSIM | SAM | SCC |
| Hysure | 28.6162 | 0.9106 | 14.1646 | 0.6066 | 26.0198 | 0.8629 | 17.7089 | 0.4302 | 22.2750 | 0.8378 | 15.3534 | 0.2962 |
| FUSE | 27.8624 | 0.9269 | 12.7784 | 0.5856 | 27.6576 | 0.9202 | 13.4699 | 0.5268 | 26.7657 | 0.9083 | 13.4097 | 0.3370 |
| GSA | 26.7430 | 0.8526 | 17.7414 | 0.6566 | 24.1530 | 0.8019 | 21.5064 | 0.5885 | 23.0795 | 0.7957 | 18.6764 | 0.4125 |
| Fuvar | 27.8725 | 0.7143 | 11.6015 | 0.5488 | 23.4275 | 0.8362 | 19.1554 | 0.3133 | 23.8808 | 0.9358 | 9.6115 | 0.5037 |
| SSR-Net | 34.2657 | 0.9591 | 6.7539 | 0.7963 | 32.7943 | 0.9590 | 9.0995 | 0.7775 | 31.0422 | 0.9529 | 9.3713 | 0.6837 |
| ResTFNet | 34.2637 | 0.9665 | 6.3812 | 0.7766 | 32.8670 | 0.9667 | 6.5213 | 0.7446 | 27.7787 | 0.9137 | 8.3283 | 0.4016 |
| MHF-Net | 34.6613 | 0.9641 | 6.2217 | 0.8019 | 33.2460 | 0.9588 | 6.3281 | 0.7882 | 31.1512 | 0.9536 | 6.6925 | 0.6951 |
| HSRnet | 34.9831 | 0.9755 | 5.6309 | 0.7857 | 32.1052 | 0.9498 | 6.4967 | 0.6789 | 31.2970 | 0.9535 | 7.3475 | 0.6575 |
| FTDN | **37.0362** | **0.9811** | **4.9171** | **0.8544** | **36.0897** | **0.9796** | **5.0092** | **0.8338** | **33.7255** | **0.9749** | **5.4581** | **0.7632** |

* The ideal values of PSNR, SSIM, SAM and SCC are $+\infty$, 1, 0 and 1, respectively.

TABLE III

QUANTITATIVE INDICATORS OF IMAGE FUSION OF CASE1 ON THE DY AND YRE DATASETS. (THE BEST IS MARKED IN BOLD AND THE SECOND BEST IS UNDERLINED)

| | DY dataset | | | | | | | YRE dataset | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Methods | PSNR | SSIM | SAM | SCC | Ds | $D_\lambda$ | QNR | PSNR | SSIM | SAM | SCC | Ds | $D_\lambda$ | QNR |
| Hysure | 28.6162 | 0.9106 | 14.1464 | 0.6066 | 0.0866 | 0.0352 | 0.8813 | 23.7100 | 0.7948 | 3.0335 | 0.5086 | 0.0866 | 0.0352 | 0.8813 |
| FUSE | 27.8624 | 0.9269 | 12.7784 | 0.5856 | **0.0019** | 0.0509 | 0.9492 | 25.5382 | 0.7494 | 3.4222 | 0.5191 | 0.1500 | 0.1642 | 0.7102 |
| GSA | 26.7430 | 0.8526 | 17.7414 | 0.6566 | 0.0037 | 0.1037 | 0.8930 | 24.4379 | 0.7498 | 3.1730 | 0.5496 | 0.0639 | 0.3311 | 0.6261 |
| Fuvar | 27.8725 | 0.7143 | 11.6015 | 0.5488 | - | - | - | 25.4443 | 0.7071 | 4.0109 | 0.6277 | - | - | - |
| SSR-Net | 34.2657 | 0.9591 | 6.7539 | 0.7963 | 0.0032 | - | - | 30.7925 | 0.8740 | 2.2858 | 0.8288 | 0.1203 | 0.0489 | 0.8366 |
| ResTFNet | 34.2637 | 0.9665 | 6.3812 | 0.7766 | 0.0311 | 0.0211 | 0.9485 | 31.6507 | 0.8840 | 2.1335 | 0.8451 | 0.0290 | 0.0480 | 0.9244 |
| MHF-Net | 34.6613 | 0.9641 | 6.2217 | 0.8019 | 0.0030 | 0.0300 | 0.9672 | 29.2452 | 0.8599 | 3.0028 | 0.8305 | 0.0435 | 0.3331 | 0.6379 |
| HSRnet | 34.9831 | 0.9755 | 5.6309 | 0.7857 | 0.0031 | **0.0185** | **0.9785** | 29.1432 | 0.8594 | 2.4778 | 0.7597 | **0.0208** | 0.0294 | **0.9505** |
| FTDN | **37.0362** | **0.9811** | **4.9171** | **0.8544** | 0.0035 | 0.0447 | 0.9520 | **33.1906** | **0.9054** | **1.7158** | **0.8726** | 0.0494 | **0.0214** | 0.9303 |

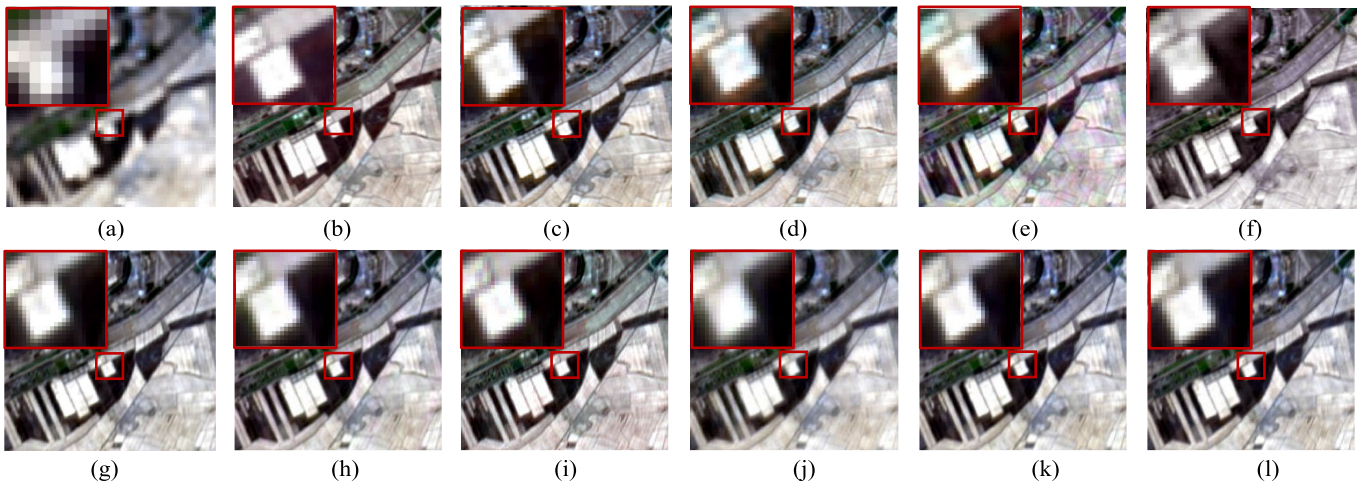


Fig. 3. Visual comparison of the fused results on the YRE dataset for Case 1. (a) Reduced LR–HS image. (b) Reduced HR–MS image. (c)–(k) Fusion results of different methods displayed in pseudo-color, with a demarcated area zoomed in the upper left corner. (l) Reference of HR-HS images.

with varying degrees and our method is objectively leading among all the methods. Fig. 4 depicts that incompatible surface features cause slight chromatic aberration in the results of traditional methods. Our proposed method restores the real information, including a finer spectrum and realistic structure. Other DL-based methods miss some details during the fusion.

*3) Case 3:* Temporal difference of more than a year causes widespread land cover changes and phenological varieties, which is hard to exploit the complementary information from

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:05:10 UTC from IEEE Xplore. Restrictions apply.

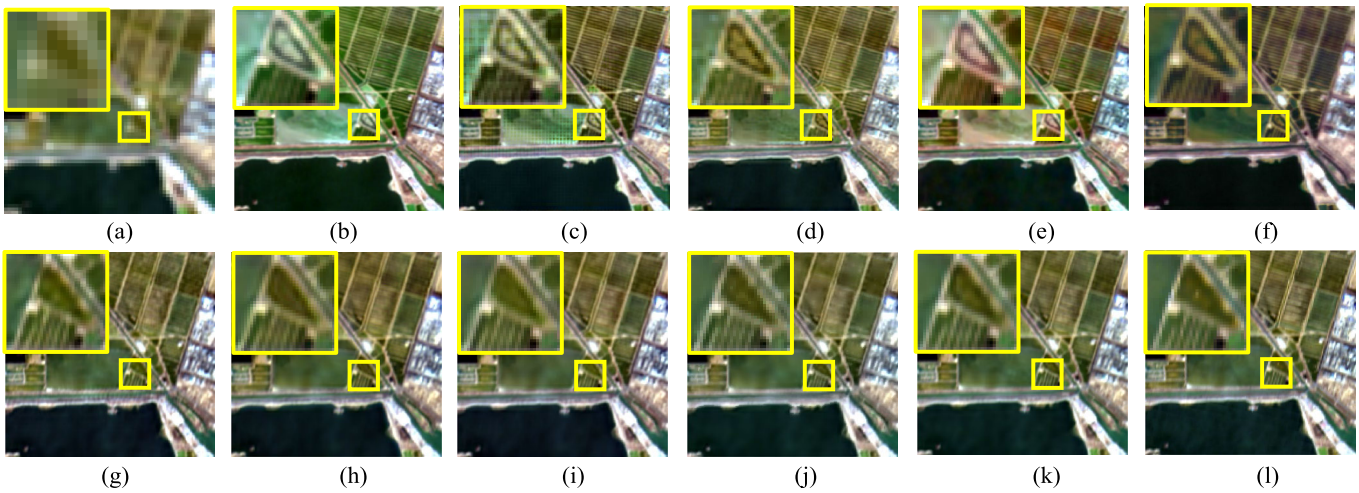



Fig. 4. Visual comparison of the fused results on the DY dataset for Case 2. (a) Original LR–HS image. (b) Original HR–MS image. (c)–(k) Fusion results of different methods displayed in pseudo-color, with a demarcated area zoomed in the upper left corner. (l) Reference of HR-HS images.

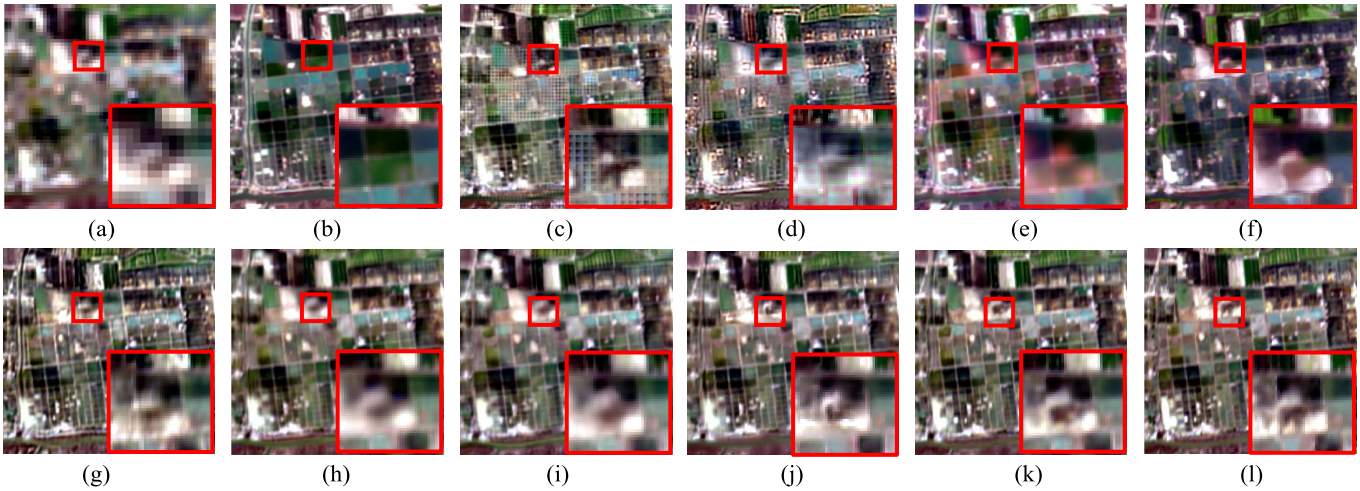



Fig. 5. Visual comparison of the fused results on the DY dataset for Case 3. (a) Original LR-HS image. (b) Original HR–MS image. (c)–(k) Fusion results of different methods displayed in pseudo-color, with a demarcated area zoomed in the lower right corner. (l) Reference of HR-HS images.

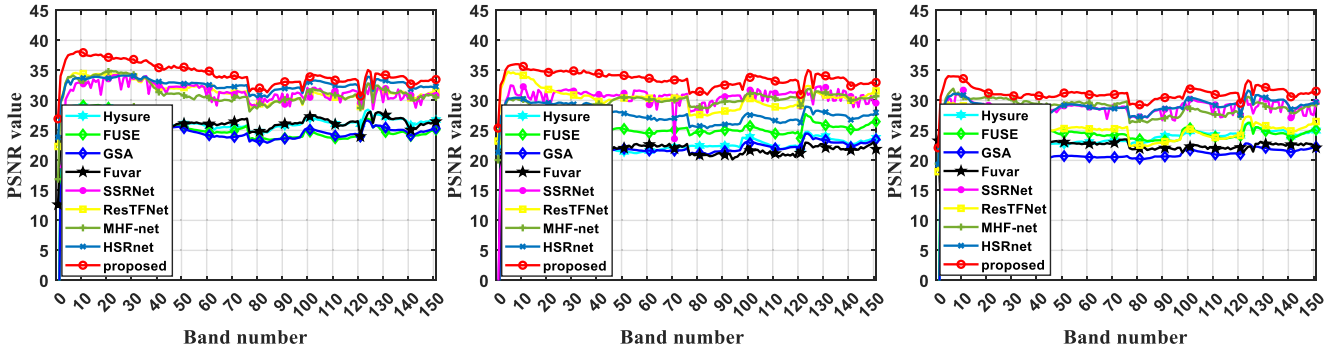



Fig. 6. Curves of PSNR values covering all spectral bands on the DY dataset. (a) Case 1. (b) Case 2. (c) Case 3.

corresponding areas. The indicators in the rightmost of Table II are shown that PSNR and SCC fall off a cliff, and both the traditional and DL-based methods performed mediocrely. The visual results are shown in Fig. 5, and irregular changes in land covers are reflected in the zoomed images, GSA cannot handle spectral distortion, and other methods are confused by undesirable spatial artifacts. Hysure suffers from severe grid noise. For this case, our method gives a good solution for the deep excavation of unmatched information. Moreover, we also present the corresponding SAM error maps of comparison methods in Fig. 6 for intuitive comparison. The closer the color is to the reference, the better the spectral quality is. Judging the maps from Fig. 6(a)–(j), distortion regions in yellows are plainly less in our error map.

Eventually, to determine the individual spectral band quality of each fused image for the three cases, the curves of PSNR covering all spectral bands are displayed in Fig. 7. As can be seen, the proposed method achieves the best reconstruction

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:05:10 UTC from IEEE Xplore. Restrictions apply.

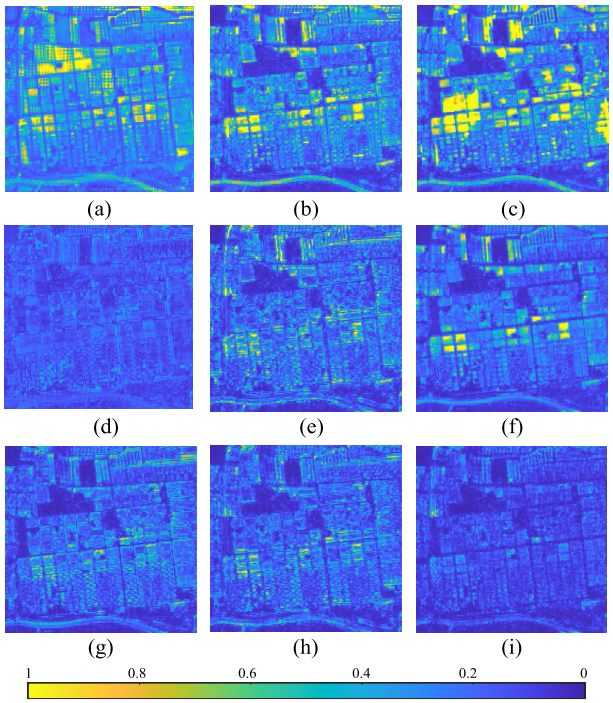



Fig. 7. SAM error maps of the comparison methods on the DY dataset of Case 3. (a)–(i) Fusion results of different methods.

quality in various bands and this advantage is particularly apparent when the temporal difference span accretes.

In summary, our model can generate a more reliable fusion result in various temporal difference spans and behaves less susceptible reflected in the unstable regions.

### C. Comparison on Full-Resolution Scene

In this section, full-resolution experiments are conducted on the DY and YRE datasets as supplements to reduced-resolution experiments and further validate the real-world generalization abilities. To be noted, there are no appropriate evaluation means for Cases 2 and 3, because of the mismatched input data.

*1) Case 1:* Table III shows quantitative evaluations of the DY and YRE datasets. QNR is composed of $D_s$ and $D_\lambda$, which measure the distortions in texture and spectrum, respectively. It can be seen that DL-based methods are generally superior to traditional methods, and the proposed methods achieve competitive results compared with other methods. The visualization of the YRE dataset is shown in Fig. 8, where Fig. 8(a) and (b) shows the original HR-MS and LR-HS images as spatial and spectral references, respectively. On the second line, SAM error maps highlight the regions of lack spectral fidelity in the reconstructed images, and all the results need to be downsampled to the same size as the reference LR-HS image. Some methods suffer from spatial blur, that is, HSRnet, while some methods suffer from spectral distortion, that is, Hysure and SSRNet. In contrast, our method can well exploit complementary information and receive a balanced fidelity effect.

*2) Case 2:* Fig. 9 displays the visual results of Case 2. In contrast with Fig. 9(a) and (b), some spectral changes occurred in the water and vegetation. Hysure is unable to handle the abnormal changes of endmember vectors and almost completely loses spectral information, while GSA obviously loses LR-HS information. As can be seen in SAM maps, the proposed method and HSRnet can minimize the spectral differences, but HSRnet causes negative artifacts in the sharp edge.

*3) Case 3:* Fig. 10 displays the visual results of Case 3. Long acquisition time interval is reflected in the wide range change of surface coverage. Traditional methods are ahead on reserved spatial details and make no response to the mutation area on the right. DL-based methods focus on the compromise between two modalities but have obstacles in the prediction of complex objects. Relatively speaking, our method performs greater advantages in the face of this challenging integration.

TABLE IV
ABLATION STUDIES OF SCM ON THE DY DATASET. (THE BEST IS MARKED IN **BOLD** LETTERS)

| # | settting | PSNR | SSIM | SAM | SCC |
|---|---|---|---|---|---|
| 1 | 3×3 convolution | 36.9218 | 0.9807 | 4.9379 | 0.8509 |
| 2 | SCM | **37.0362** | **0.9811** | **4.9171** | **0.8544** |

TABLE V
ABLATION STUDIES OF SFA ON THE DY DATASET. (THE BEST IS MARKED IN **BOLD** LETTERS)

| # | settting | PSNR | SSIM | SAM | SCC |
|---|---|---|---|---|---|
| 1 | 3×3 convolution | 36.8702 | 0.9808 | 4.9389 | 0.8535 |
| 2 | SFA | **37.0362** | **0.9811** | **4.9171** | **0.8544** |

TABLE VI
ABLATION STUDIES OF IRA ON THE DY DATASET. (THE BEST IS MARKED IN **BOLD** LETTERS)

| # | settting | PSNR | SSIM | SAM | SCC |
|---|---|---|---|---|---|
| 1 | Long-skip connection | 36.8244 | 0.9809 | **4.8987** | 0.8525 |
| 2 | IRA | **37.0362** | **0.9811** | 4.9171 | **0.8544** |

TABLE VII
ABLATION STUDIES OF THE NUMBER OF MHCIL ON THE DY DATASET. (THE BEST IS MARKED IN **BOLD** LETTERS)

| # | settting | PSNR | SSIM | SAM | SCC |
|---|---|---|---|---|---|
| 1 | N=1 | 36.8702 | 0.9805 | 4.9772 | 0.8535 |
| 2 | N=2 | 36.8862 | 0.9807 | **4.9056** | 0.8534 |
| 3 | N=3 | **37.0362** | **0.9811** | 4.9171 | **0.8544** |
| 4 | N=4 | 36.8326 | **0.9811** | 4.9300 | 0.8537 |

TABLE VIII
RUNNING TIME OF COMPARISON METHOD ON THE DY TEST SET. (THE BEST IS MARKED IN **BOLD** AND THE SECOND BEST IS UNDERLINED)

| Method | Reduced-resolution(s) | Full-resolution(s) |
|---|---|---|
| Hysure | 29.3843(CPU) | 292.5933(CPU) |
| FUSE | 6.0883(CPU) | 7.1245(CPU) |
| GSA | 3.0361(CPU) | 7.6826(CPU) |
| Fuvar | 729.4196(CPU) | - |
| ResTFNet | 0.0045(GPU) | 0.0162(GPU) |
| SSRNet | **0.0030**(GPU) | **0.0157**(GPU) |
| MHF-Net | 0.4142(GPU) | 2.2476(GPU) |
| HSRnet | 0.0587(GPU) | 0.1795(GPU) |
| FTDN | 0.4098(GPU) | 2.1847(GPU) |

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:05:10 UTC from IEEE Xplore. Restrictions apply.

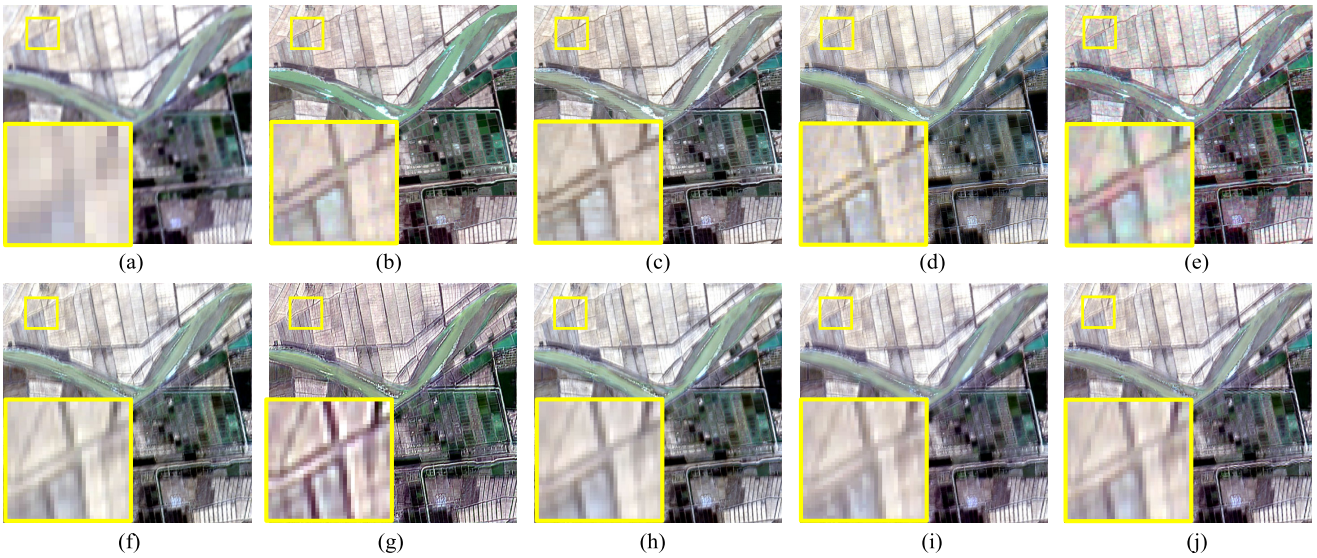



Fig. 8. Visual comparison of the fused results on the YRE real dataset for Case 1. (a) Original LR-HS image. (b) Original HR-MS image. (c)–(j) Fusion results of different methods.

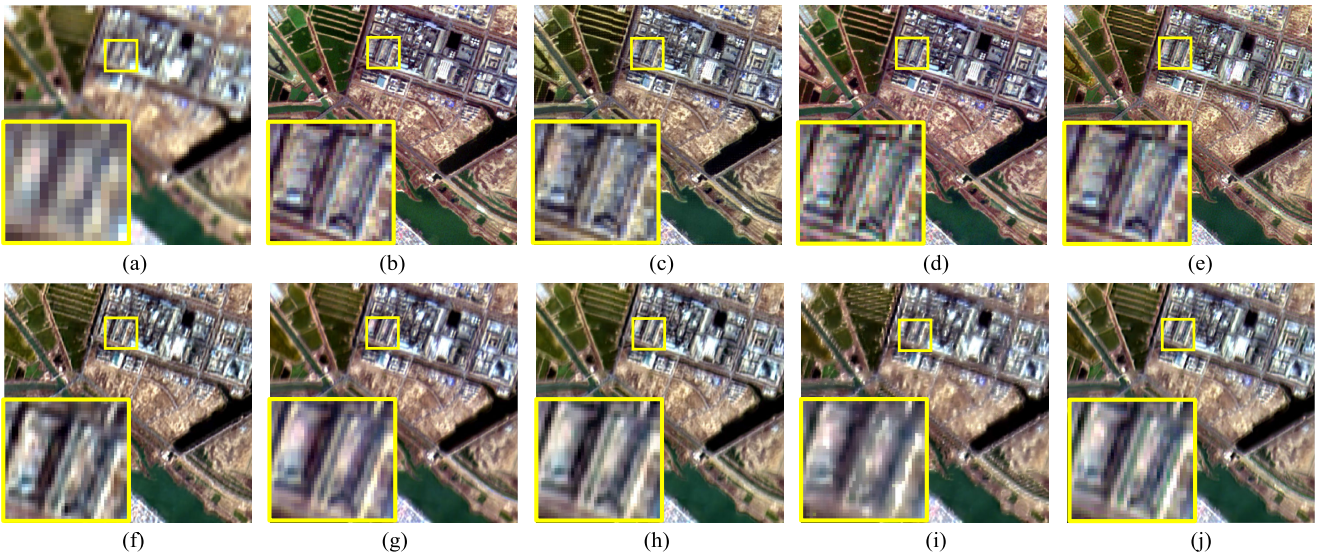



Fig. 9. Visual comparison of the fused results on the DY real dataset for Case 2. (a) Original LR-HS image. (b) Original HR-MS image. (c)–(j) Fusion results of different methods.

TABLE IX
SPECIFIC CONTENT OF THE PAVIA_U DATASET

| Pavia_U dataset | | | | | |
|---|---|---|---|---|---|
| Scale factor | Reference HR-HS | LR-HS | HR-MS | Testing LR-HS | Testing HR-MS |
| 4 | 600×340×103 | 150×85×103 | 600×340×4 | 45×45×103 | 180×180×4 |
| 8 | 600×336×103 | 75×42×103 | 600×336×4 | 25×25×103 | 200×200×4 |

### D. Ablation Studies

In this section, we conduct ablation studies of FTDN on the DY dataset for Case 1 to further verify the worth of the designed components, including SCM, SFA, and the number of MHCIL. Tables IV–VII illustrate the average results on the testing dataset.

*1) Effectiveness of SCM:* The proposed SCM utilizes spectral correlation to discover the common information inter features. To show the effect of SCM, we removed it and discussed an alternative of concatenation with a 3 × 3 convolution block for feature interaction (termed as w/o SCM). The results shown in the first group of Table IV verify that the spectral correlation-based strategy is decisive when substituting it causes obvious performance degradation in PSNR and SSIM, and simple integration cannot offer sufficient support.

*2) Effectiveness of SFA:* In the second group of Table V, we removed a couple of SFA from the network instead of concatenation with a 3 × 3 convolution block and kept the remaining structure unchanged (termed as w/o SFA). As can

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:05:10 UTC from IEEE Xplore. Restrictions apply.




Fig. 10. Visual comparison of the fused results on the DY real dataset for Case 3. (a) Original LR-HS image. (b) Original HR-MS image. (c)–(j) Fusion results of different methods.

be seen, PSNR and SAM suffer a decline in the ablation experiment.

*3) Effectiveness of IAR:* We removed image adaptive reconstruction (IAR) operation from the network, instead of adding $F_o$ to the original HS ↑ through a long-skip connection (termed as w/o IAR). The third group of Table VI shows that IAR provides a better solution for feature reconstruction, especially in spatial edge-preserving.

*4) Effectiveness of the Number of MHCIL:* We change the number of MHCIL to pursue a good compromise between parameters and performance, and the interactions are set to {1, 2, 3, 4}. The last group of Table VII proves that the proposed network gets the best performance when $N$ is set to 4. Thus, we set $N$ to 4 throughout the studies.

Taken together, the effectiveness of the above-mentioned components has been verified and become the basis for the design of the proposed model.

### E. Computational Efficiency

In this section, we verified the computational efficiency of different methods on the reduced-resolution and full-resolution DY test sets. The traditional method was executed on the CPU, while the DL-based methods were executed on the GPU. Table VIII shows the average running time of the comparison methods. As can be seen that SSR-Net has a lower demand for trainable parameters, and the efficiency of the DL-based methods using GPU is far better than traditional methods.

### F. Discussion

*1) Generalizability to the Public Dataset:* To verify the generalizability of the proposed method on public datasets, we conducted experiments on the Pavia University (Pavia_U) dataset. The experimental setup is shown in Table IX, with a spatial degradation factor of 4 and a 180 × 180 image patch cropped for testing. The experimental results are shown in Table X and Fig. 11, and the proposed method has the best performance, with more restored details.

TABLE X

QUANTITATIVE INDICATORS OF IMAGE FUSION ON THE PAVIA_U DATASET. (THE BEST IS MARKED IN **BOLD** AND THE SECOND BEST IS UNDERLINED)

| Pavia_U dataset | | | | | |
|---|---|---|---|---|---|
| Methods | Scale factor | PSNR | SSIM | SAM | SCC |
| Hysure | ×4 | 40.7462 | 0.9965 | 2.6260 | 0.9662 |
| FUSE | | 35.7754 | 0.9744 | 3.0955 | 0.9670 |
| GSA | | 35.7247 | 0.9903 | 3.3077 | 0.9554 |
| ResTFNet | | 40.6209 | 0.9930 | 2.6094 | 0.9424 |
| SSR-Net | | 41.6239 | 0.9986 | 2.4341 | 0.9647 |
| MHF-Net | | 42.5346 | 0.9982 | 2.3094 | 0.9635 |
| HSRnet | | 42.8628 | 0.9979 | 2.2181 | 0.9646 |
| FTDN | | **43.5315** | **0.9991** | **2.1470** | **0.9697** |
| Hysure | ×8 | 37.7408 | 0.9920 | 3.3598 | 0.9577 |
| FUSE | | 35.6105 | 0.9737 | 4.5404 | **0.9585** |
| GSA | | 33.2060 | 0.9805 | 4.8907 | 0.9275 |
| ResTFNet | | 38.5149 | 0.9935 | 3.0072 | 0.9305 |
| SSR-Net | | 41.0013 | **0.9977** | **2.5570** | 0.9538 |
| MHF-Net | | 40.7216 | 0.9976 | 2.6561 | 0.9504 |
| HSRnet | | 38.3515 | 0.9851 | 2.8448 | 0.9062 |
| FTDN | | **41.0427** | 0.9941 | 2.8961 | 0.9551 |

*2) Generalizability to Different Scale:* In addition, to verify the performance under different scale factors, we conducted experiments with a spatial degradation factor of 8 and a 200 × 200 image patch cropped for testing. As shown in Table X and Fig. 12, the proposed method achieved the best values on PSNR but showed a decline in spectral fidelity. The reason for this, in addition to overfitting, should be that large-scale factors are not conducive to accurate weight allocation in SCM, which is also a part that needs further improvement.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:05:10 UTC from IEEE Xplore. Restrictions apply.

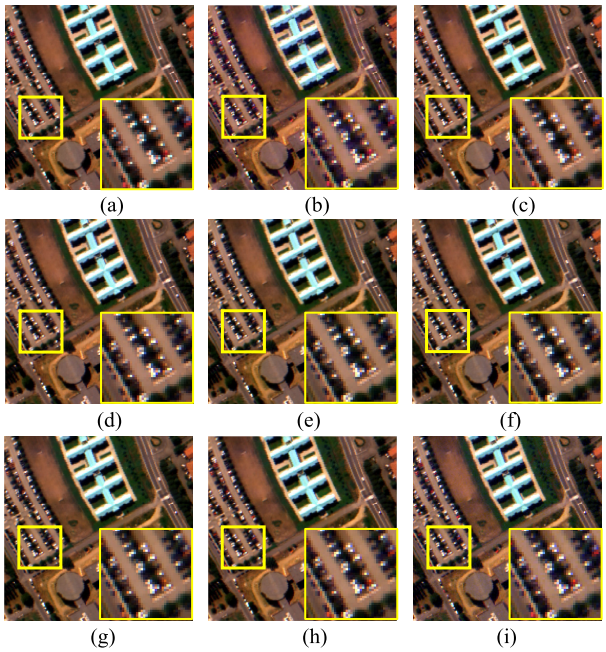



Fig. 11. Visual comparison of the fused results on the Pavia_U dataset with a scale factor of 4 for Case 1. (a)–(h) Fusion results of different methods displayed in pseudo-color, with a demarcated area zoomed in the lower right corner. (i) Reference of HR-HS images.

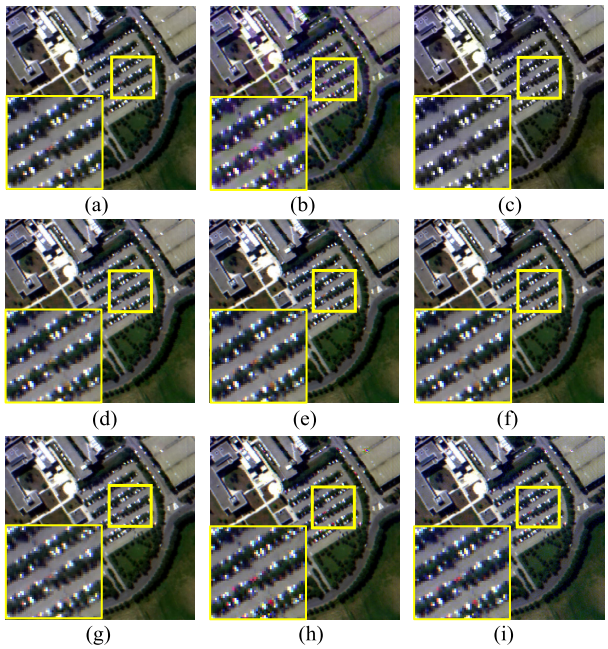



Fig. 12. Visual comparison of the fused results on the Pavia_U dataset with a scale factor of 8 for Case 1. (a)–(h) Fusion results of different methods displayed in pseudo-color, with a demarcated area zoomed in the lower left corner. (i) Reference of HR-HS images.

Combining all experiments, the existing MS/HS fusion methods are in a dilemma when imaging conditions are inconsistent, for example, Cases 2 and 3. The traditional methods regard the observations to be equivalent when performing lineal feature processing, struggling in the distorted region. Among them, optimization-based methods such as Fuvar and Hysure, due to imperfect spectral overlap, are apparently better in terms of spectral quality than the CS-based method GSA, while GSA outperforms in spatial details. The poor visualization of FUSE is believed to come from defective subspace representation.

Deep-learning-based methods, benefiting from nonlinear characterization, show significant advantages in land cover reconstruction. However, the existing networks center on the early fusion structure, which merges the redundant information in the initial stage, resulting in the pollution of invalid features. As can be seen above, the results of Fig. 5(g)–(k) gain better spectral qualities, but they are still not ideal for land cover changes.

The proposed FTDN, based on a matching-fusion strategy, is more competitive in rebuilding the temporal changes and shows robustness to different scenarios. However, there are still some issues that need improvement: first, we should design a stricter feature-matching module to maximize the mining and utilization of complementary information. Second, lightweight networks should be designed for MS/HS image fusion implementation.

## V. Conclusion

In this article, we propose a spectral correlation-based strategy for MS and HS image fusion with diverse temporal difference spans. Equipped with an SCM module, the proposed method exploits the common information to relieve the mismatches between multisource images caused by seasonal and land cover changes. Besides, we developed the SFA module to promote complementary feature coupling inferring from covariance. Moreover, we introduce a learning gating to motivate features to reconstruct competitively and can improve the effect at a paltry cost. Extensive experiments conducted on the DY and YRE datasets verify that our method achieves superior performance over the existing methods visually and quantitatively. Furthermore, our method is robust to diverse temporal difference spans including the same acquisition time.

## References

[1] X. Li, Z. Du, Y. Huang, and Z. Tan, "A deep translation (GAN) based change detection network for optical and SAR remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 179, pp. 14–34, Sep. 2021.

[2] M. Wan, X. Ye, X. Zhang, Y. Xu, G. Gu, and Q. Chen, "Infrared small target tracking via Gaussian curvature-based compressive convolution feature extraction," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[3] K. Xu, H. Huang, P. Deng, and Y. Li, "Deep feature aggregation framework driven by graph convolutional network for scene classification in remote sensing," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 10, pp. 5751–5765, Oct. 2022.

[4] F. Laporterie-Déjean, H. de Boissezon, G. Flouzat, and M.-J. Lefèvre-Fonollosa, "Thematic and statistical evaluations of five panchromatic/multispectral fusion methods on simulated PLEIADES-HR images," *Inf. Fusion*, vol. 6, no. 3, pp. 193–212, 2005.

[5] H. Shen, X. Meng, and L. Zhang, "An integrated framework for the spatio-temporal–spectral fusion of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7135–7148, Dec. 2016.

[6] J. Li, Y. Li, L. He, J. Chen, and A. Plaza, "Spatio-temporal fusion for remote sensing data: An overview and new benchmark," *Sci. China Inf. Sci.*, vol. 63, no. 4, pp. 1–17, Apr. 2020.

[7] R. A. Borsoi, T. Imbiriba, and J. C. M. Bermudez, "Super-resolution for hyperspectral and multispectral image fusion accounting for seasonal spectral variability," *IEEE Trans. Image Process.*, vol. 29, pp. 116–127, 2020.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:05:10 UTC from IEEE Xplore. Restrictions apply.

[8] R. A. Borsoi, C. Prévost, K. Usevich, D. Brie, J. C. M. Bermudez, and C. Richard, "Coupled tensor decomposition for hyperspectral and multispectral image fusion with inter-image variability," *IEEE J. Sel. Topics Signal Process.*, vol. 15, no. 3, pp. 702–717, Apr. 2021, doi: 10.1109/JSTSP.2021.3054338.

[9] D. Wang, Y. Bai, C. Wu, Y. Li, C. Shang, and Q. Shen, "Convolutional LSTM-based hierarchical feature fusion for multispectral pan-sharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5404016.

[10] Y. Yang, C. Wang, Y. Feng, J. Zhang, Y. Zheng, and S. Chen, "Regularizing subspace representation for fusing hyperspectral and multispectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 12273–12286, 2021.

[11] S. Takeyama and S. Ono, "Robust hyperspectral image fusion with simultaneous guide image denoising via constrained convex optimization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5544518.

[12] N. Liu, W. Li, Y. Wang, R. Tao, Q. Du, and J. Chanussot, "A survey on hyperspectral image restoration: From the view of low-rank tensor approximation," *Sci. China Inf. Sci.*, vol. 66, no. 4, Apr. 2023, Art. no. 140302.

[13] T. Gelvez-Barrera, H. Arguello, and A. Foi, "Joint nonlocal, spectral, and similarity low-rank priors for hyperspectral—Multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5537112, doi: 10.1109/TGRS.2022.3203294.

[14] C. Prévost, R. A. Borsoi, K. Usevich, D. Brie, J. C. M. Bermudez, and C. Richard, "Hyperspectral super-resolution accounting for spectral variability: Coupled tensor LL1-based recovery and blind unmixing of the unknown super-resolution image," *SIAM J. Imag. Sci.*, vol. 15, no. 1, pp. 110–138, Mar. 2022.

[15] X. Meng, H. Shen, H. Li, L. Zhang, and R. Fu, "Review of the pansharpening methods for remote sensing images based on the idea of meta-analysis: Practical discussion and challenges," *Inf. Fusion*, vol. 46, pp. 102–113, Mar. 2019.

[16] F. Palsson, J. R. Sveinsson, M. O. Ulfarsson, and J. A. Benediktsson, "Model-based fusion of multi- and hyperspectral images using PCA and wavelets," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 5, pp. 2652–2663, May 2015.

[17] B. Aiazzi, L. Alparone, S. Baronti, and M. Selva, "MS + Pan image fusion by enhanced Gram-Schmidt spectral sharpening," in *Proc. 26th EARSel Symp.*, 2006, pp. 1–8.

[18] B. Aiazzi, L. Alparone, S. Baronti, A. Garzelli, and M. Selva, "MTF-tailored multiscale fusion of high-resolution MS and pan imagery," *Photogramm. Eng. Remote Sens.*, vol. 72, no. 5, pp. 591–596, May 2006.

[19] X. Otazu, M. Gonzalez-Audicana, O. Fors, and J. Nunez, "Introduction of sensor spectral response into image fusion methods. Application to wavelet-based methods," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 10, pp. 2376–2385, Oct. 2005.

[20] F. Fang, F. Li, C. Shen, and G. Zhang, "A variational approach for pan-sharpening," *IEEE Trans. Image Process.*, vol. 22, no. 7, pp. 2822–2834, Jul. 2013.

[21] C. Chen, Y. Li, W. Liu, and J. Huang, "SIRF: Simultaneous satellite image registration and fusion in a unified framework," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 4213–4224, Nov. 2015.

[22] V. F. Haertel and Y. E. Shimabukuro, "Spectral linear mixing model in low spatial resolution image data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 11, pp. 2555–2562, Nov. 2005.

[23] M. Simoes, J. Bioucas-Dias, L. B. Almeida, and J. Chanussot, "A convex formulation for hyperspectral image superresolution via subspace-based regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 3373–3388, Jun. 2015.

[24] N. Yokoya, T. Yairi, and A. Iwasaki, "Coupled nonnegative matrix factorization unmixing for hyperspectral and multispectral data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 528–537, Feb. 2012.

[25] Z. Wang, B. Chen, R. Lu, H. Zhang, H. Liu, and P. K. Varshney, "FusionNet: An unsupervised convolutional variational network for hyperspectral and multispectral image fusion," *IEEE Trans. Image Process.*, vol. 29, pp. 7565–7577, 2020.

[26] R. Dian, S. Li, and X. Kang, "Regularizing hyperspectral and multispectral image fusion by CNN denoiser," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 3, pp. 1124–1135, Mar. 2021.

[27] T. Uezato, D. Hong, N. Yokoya, and W. He, "Guided deep decoder: Unsupervised image pair fusion," in *Proc. ECCV*, 2020, pp. 87–102.

[28] X. Meng, N. Wang, F. Shao, and S. Li, "Vision transformer for pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5409011.

[29] G. Masi, D. Cozzolino, L. Verdoliva, and G. Scarpa, "Pansharpening by convolutional neural networks," *Remote Sens.*, vol. 8, no. 7, p. 594, Jul. 2016.

[30] Q. Xie, M. Zhou, Q. Zhao, Z. Xu, and D. Meng, "MHF-Net: An interpretable deep network for multispectral and hyperspectral image fusion," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 3, pp. 1457–1473, Mar. 2022.

[31] X. Zhang, W. Huang, Q. Wang, and X. Li, "SSR-NET: Spatial–spectral reconstruction network for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5953–5965, Jul. 2021.

[32] J.-F. Hu, T.-Z. Huang, L.-J. Deng, T.-X. Jiang, G. Vivone, and J. Chanussot, "Hyperspectral image super-resolution via deep spatiospectral attention convolutional neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 12, pp. 7251–7265, Dec. 2022.

[33] C. Zhao, X. Gao, W. J. Emery, Y. Wang, and J. Li, "An integrated spatio-spectral–temporal sparse representation method for fusing remote-sensing images with different resolutions," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3358–3370, Jun. 2018, doi: 10.1109/TGRS.2018.2798663.

[34] Y. Peng, W. Li, X. Luo, J. Du, Y. Gan, and X. Gao, "Integrated fusion framework based on semicoupled sparse tensor factorization for spatio-temporal–spectral fusion of remote sensing images," *Inf. Fusion*, vol. 65, pp. 21–36, Jan. 2021.

[35] J. Zhou, W. Sun, X. Meng, G. Yang, K. Ren, and J. Peng, "Generalized linear spectral mixing model for spatial–temporal–spectral fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5533216, doi: 10.1109/TGRS.2022.3188501.

[36] Q. Liu, X. Chen, X. Meng, H. Chen, F. Shao, and W. Sun, "Dual-task interactive learning for unsupervised spatio-temporal–spectral fusion of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5402015, doi: 10.1109/TGRS.2023.3265400.

[37] W. Sun, K. Ren, X. Meng, C. Xiao, G. Yang, and J. Peng, "A band divide-and-conquer multispectral and hyperspectral image fusion method," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5502113.

[38] F. Yang, H. Yang, J. Fu, H. Lu, and B. Guo, "Learning texture transformer network for image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 5791–5800.

[39] M. Hassanin, S. Anwar, I. Radwan, F. S. Khan, and A. Mian, "Visual attention methods in deep learning: An in-depth survey," 2022, *arXiv:2204.07756*.

[40] L. Wald, T. Ranchin, and M. Mangolini, "Fusion of satellite images of different spatial resolutions: Assessing the quality of resulting images," *Photogramm. Eng. Remote Sens.*, vol. 63, no. 6, pp. 691–699, 1997.

[41] Q. Wei, N. Dobigeon, and J.-Y. Tourneret, "Fast fusion of multi-band images based on solving a Sylvester equation," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 4109–4121, Nov. 2015.

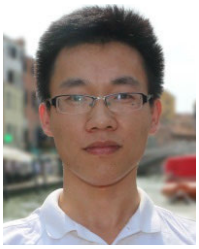
[42] X. Liu, Q. Liu, and Y. Wang, "Remote sensing image fusion based on two-stream fusion network," *Inf. Fusion*, vol. 55, pp. 1–15, Mar. 2020.



**Xu Chen** received the B.S. degree from Ningbo University, Ningbo, China, in 2021, where he is currently pursuing the M.S. degree.

His research interests include hyperspectral image processing and multisource data fusion.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:05:10 UTC from IEEE Xplore. Restrictions apply.



**Xiangchao Meng** (Member, IEEE) received the B.S. degree in geographic information systems from the Shandong University of Science and Technology, Qingdao, China, in 2012, and the Ph.D. degree in cartography and geography information systems from Wuhan University, Wuhan, China, in 2017.

He is currently an Associate Professor with the Faculty Member of Electrical Engineering and Computer Science, Ningbo University, Ningbo, China. His research interests include multisource data fusion and machine learning.



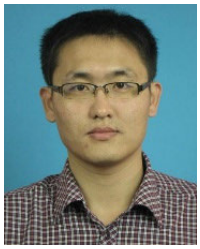
**Qiang Liu** received the B.S. degree from Shenyang Ligong University, Shenyang, China, in 2016, and the M.S. degree from the School of Ningbo University, Ningbo, China, in 2019. He is currently pursuing the Ph.D. degree with Ningbo University, Ningbo.

His research interests include deep learning for image processing, artificial intelligence, and multisource image fusion.



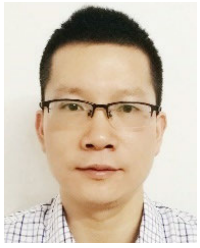
**Huiping Jiang** received the Ph.D. degree in cartography and geographic information system from the University of Chinese Academy of Sciences, Beijing, China, in 2019.

Following his three-year post-doctoral study, he has been with the Key Laboratory of Regional Sustainable Development Modeling, Chinese Academy of Sciences, Beijing, as an Assistant Research Fellow, since 2023. His research interests include urban remote sensing, big data for SDG 11, sustainable urbanization, and human dimensions of global environmental change.



**Gang Yang** (Member, IEEE) received the M.S. degree in geographical information systems from the Hunan University of Science and Technology, Xiangtan, China, in 2012, and the Ph.D. degree from the School of Resource and Environmental Sciences, Wuhan University, Wuhan, China, in 2016.

He is currently an Associate Professor with Ningbo University, Ningbo, Zhejiang, China. His research interests include missing information reconstruction of remote-sensing images, cloud removal of remote-sensing images, and remote-sensing time-series products temporal reconstruction.



**Weiwei Sun** (Senior Member, IEEE) received the B.S. degree in surveying and mapping and the Ph.D. degree in cartography and geographic information engineering from Tongji University, Shanghai, China, in 2007 and 2013, respectively.

From 2011 to 2012, he was a Visiting Scholar with the Department of Applied Mathematics, University of Maryland College Park, College Park, MD, USA, with the famous Professor John Benedetto to study the dimensionality reduction of hyperspectral images. From 2014 to 2016, he was a Post-Doctoral with the State Key Laboratory for Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University, Wuhan, China, where he was involved in the study of intelligent processing in hyperspectral imagery. From 2017 to 2018, he was a Visiting Scholar in hyperspectral image processing with the Department of Electrical and Computer Engineering, Mississippi State University, Mississippi, MS, USA. He is currently a Full Professor with Ningbo University, Ningbo, Zhejiang, China. He has published more than 80 journal articles. His research interests include hyperspectral image processing with machine learning.



**Feng Shao** (Member, IEEE) received the B.S. and Ph.D. degrees in electronic science and technology from Zhejiang University, Hangzhou, China, in 2002 and 2007, respectively.

He was a Visiting Fellow with the School of Computer Engineering, Nanyang Technological University, Singapore, from 2012 to 2012. He is currently a Professor with the Faculty Member of Information Science and Engineering, Ningbo University, Ningbo, China. He has published over 100 technical articles in refereed journals and proceedings in the areas of 3-D video coding, 3-D quality assessment, and image perception.

Dr. Shao received the Excellent Young Scholar Award from the Natural Science Foundation (NSF) of China (NSFC), in 2016.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:05:10 UTC from IEEE Xplore. Restrictions apply.